

# Studying Children's Object Interaction in Virtual Reality: A Manipulative Gesture Taxonomy for VR Hand Tracking

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## ABSTRACT

In this paper, we propose a taxonomy for the classification of children's gestural input elicited from spatial puzzle play in VR hand tracking. The taxonomy builds on the existing manipulative gesture taxonomy in human-computer interaction, and offers two main analytical categories; Goal-directed actions and Hand kinematics as complementary dimensions for analysing gestural input. Based on our study with eight children (aged between 7-14), we report the qualitative results for describing the categories for analysis and quantitative results for their frequency in occurring in children's interaction with the objects during the spatial task. This taxonomy is an initial step towards capturing the complexity of manipulative gestures in relation to mental rotation actions, and helps designers and developers to understand and study children's gestures as an input for object interaction as well as an indicator for spatial thinking strategies in VR hand tracking systems.

## CCS CONCEPTS

• Human-centered computing — HCI theory, concepts and models.

## KEYWORDS

Gesture; Virtual Reality; spatial thinking; hand tracking; HCI

### ACM Reference Format:

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## 1 INTRODUCTION

Gesture is one of the powerful tools for spatial skills, which is itself inherently spatial [10]. It conveys meaning that is offered in the

language, and highlights components of an action that promote thinking and learning of abstract ideas. Hands-on interaction with physical objects is an integral part of spatial thinking skills in children [9]. With the widespread use of extended reality technologies in education and daily life [13], children's object interaction within the virtual reality (VR) contexts gain importance [2]. Thus, new forms of gesture and thought relations offered by VR systems for spatial thinking skills are yet to be explored [18, 26].

The aim of this research is to develop a taxonomy for manipulative gestures pertaining to the spatial thinking strategies of children aged 7-14 while creating geometrical objects and desired figures in VR hand tracking. We ground our understanding of gesture by using an observational study where we investigated and analysed children's gestural input in spatial puzzle play. Our preliminary results show a set of goal-directed actions elicited in different realities, and we discuss further directions for research. This paper concludes by suggesting potential avenues for future research in developing children's spatial thinking skills by utilizing VR hand tracking that incorporate analyses of manipulative gesture in studies of object interaction.

## 2 BACKGROUND

Children's interaction with physical objects is directly related to developing their spatial thinking skills [3, 9, 11, 25]. The reason for this is that physical objects contain a lot of spatial information in terms of their geometric and material properties (e.g., size, shape, location, orientation). Spatial skills are malleable and associated with the acquisition of very basic skills such as rotating and transforming objects in mind as well as finding direction in a space [21]. However, researching children's interactions with objects designed for virtual reality technologies is still a new topic, and analysing their gestural input is yet to be studied. Thus, this study provides a first step for categorizing children's gestural input in VR hand tracking systems to lie a basis for further investigations of children's mental rotation in VR.

Several studies provided gestural taxonomy in HCI such as *Quek* being one of the earliest for developing a gesture taxonomy for VR systems followed by [6, 8], gesture classifications for various multimodal interaction pertaining to gesture-based technologies (e.g. [7, 19, 24, 27]), or for interactive surfaces such as tangible tabletops (e.g. [1, 12]). Gestures can be classified in two broad categories in HCI [22] as originally inspired by the pioneering

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philological gesture taxonomy introduced by McNeill to describe the gesture and thought relation: 1) *Communicative gestures* that have an inherent communicational purpose and convey meaning that complements or supplements the accompanying speech (e.g. [12]) and 2) *Manipulative gestures* that are used to act on objects in an environment (object movement, rotation, etc.) (e.g. [23]). Inspired by literature on gesture categorization in HCI [14, 22] the taxonomy presented in this study focuses on the manipulative gestures and identifies categories specific to object interaction pertaining to mental rotation strategies. We investigate mental rotation in spatial puzzle play activity as an indicator of spatial thinking combined with hand kinematics. Recently, we have seen an instance to observe children's goal directed actions, in order to define manipulative gestures in an operational way [4]. Yet, to the best of our knowledge, there is no gesture taxonomy developed for children's object interactions in VR hand tracking systems that combine goal-directed actions and hand kinematics. Below, we further elaborate how we developed the gesture taxonomy for children's spatial object interaction in VR hand tracking.

### 3 METHOD

This study was part of a broader study that examines children's spatial thinking strategies in relation to the gestures that are used in different realities; spatial puzzle play (i.e. tangram, e.g. in [5]) in physical objects vs. in VR hand tracking. In this study, we focused particularly on developing the gesture taxonomy as a basis for our study. Thus, the relation between children's gesture input and spatial thinking strategies remains beyond the scope of this paper. In this paper, we present the method we used to develop the gesture taxonomy for children's spatial thinking in VR hand tracking.

#### 3.1 Data collection

Eight children who did not own a VR headset (2 left-handed, 5 female), aged 7-14 (Mage: 9,5) were recruited through convenience sampling, and the incentive was a gift certificate for contributing to research. One of the children was reported to have Attention Deficit and Hyperactivity Disorder and her data was discarded as she produced many off-task gestures. Thus, the final analyses include seven children in total (4 female) in this study. The data was collected with an Oculus Quest 2 VR headset that already included built-in hand-tracking sensors. The project was approved by the university ethical board. Our participants and their parents gave written consent before data collection. The play sessions took place in the VR Lab at the university campus. The parent was present in the lab during the experiment, but did not intervene. Every participant has a unique ID for this study to assure anonymity. The data was stored in a secure computer with password.

Children were asked to solve a spatial task that involved building a cat figure with tangram puzzles in VR hand tracking; the task contained two congruent sub-conditions (i.e., with gravity and without gravity). Given that it was our participants' first time using a VR headset, we first ran a warm-up session with the VR hand tracking before they started to build the tangram models. The experimenter briefly informed the participant what VR is. We also informed the participants that they could leave the study

if they feel uncomfortable or give a break whenever they want. The experimenter also informed briefly that they need to move their hands a bit slower than usual in hand tracking, otherwise the system may not track hand movements at the usual pace. The warm-up session took 3 minutes in total. We audio- and video-recorded the play sessions with two cameras; one capturing the entire body movements, whereas the other capturing the computer screen to view what the participant is seeing and doing from the VR headsets.

#### 3.2 Data analysis: Development of the Taxonomy

We analysed the audio-visual recordings by coding children's gestural input captured by two cameras. To identify gestural input in spatial puzzle play, we analyzed the hand movements in object interaction taking place in VR hand tracking. Due to the complex nature of gesture production, we coded the video data by adopting an iterative inductive coding approach [17], allowing categories to emerge from the data. Our coding workflow for data analysis involved four iterative steps:

**1st iteration - Inductive coding for determining category definitions for goal-directed actions:** The first author conducted an initial analysis based on an informal observation and sorted the goal-directed actions found in all participants' object interaction during the task.

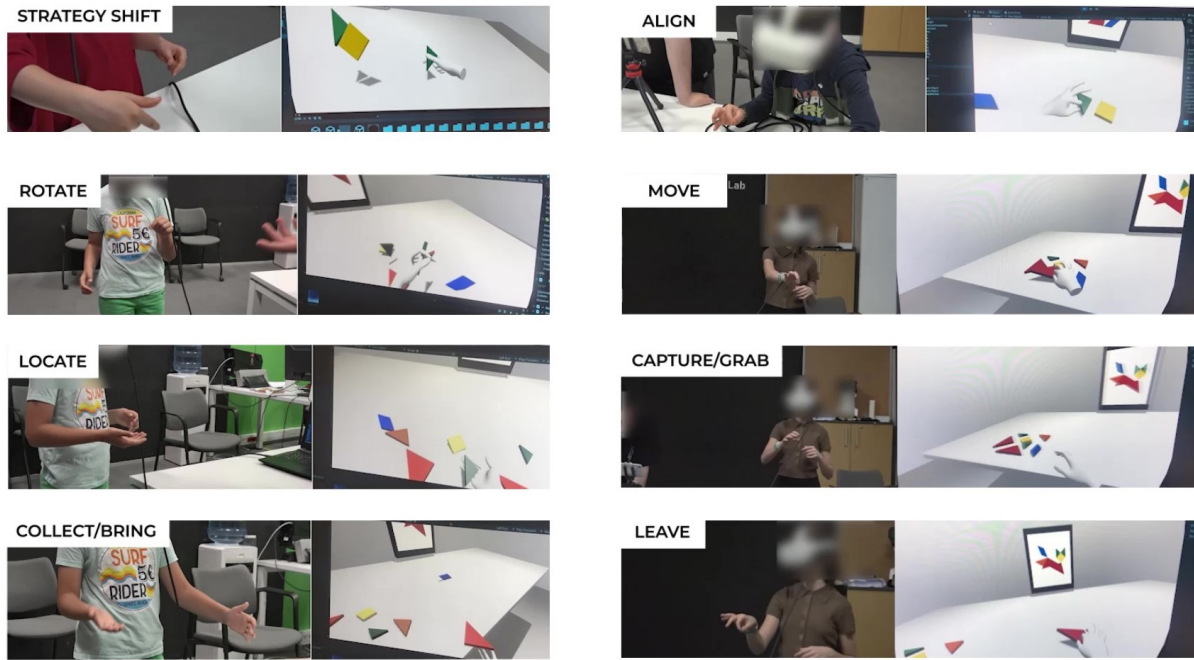
- Goal-directed Actions: Capture/grab object, Locate object, Leave object, Rotate object, Align objects, Strategy shift, and System breakdown.

The 1st iteration revealed that hand kinematics across and within goal-directed actions showed variation in hand kinematics. For a more comprehensive analysis, hand kinematics were also identified in the 2nd iteration.

**2nd iteration - A priori and inductive coding for determining category definitions for hand kinematics:** To identify the hand kinematics in children's object interaction, we partly used existing taxonomy developed by Querk who developed a gesture vocabulary based on palm orientations (e.g. up, down, right, left), and Choi et al. and McIntosh et al. who suggested categorization for fingers and hands separately (e.g. the number of hands used, the dominant vs. non-dominant hand-use, finger digits, finger pose, etc.) and combined these with self-determined kinematics that we found in our own data. Our categorization for hand kinematics included two broad sub-categories namely hand and figure configurations and we identified them with respect to their sub-categories as below:

- Hand configurations: Palm orientation (up, down, left, right), Open palm, Whole-hand grabbing.
- Finger configurations: One fingertip only (pointing), Two-finger pinch (Two-finger pinch with fistful fingers and Two-finger pinch with spread fingers), Two-finger open.

**3rd iteration - Thorough analysis of gestural input and refinement of categories through consensus coding:** After the first author identified the main categories for gesture coding, the second and the third authors coded gestures of two participants' data individually based on the defined taxonomy. We used ELAN [15] for analysing the videos that show the participant and the participant's screen view in the virtual room (see Figure 1). ELAN



**Figure 1: Goal-directed actions that occurred in children's interaction with the objects during the spatial task (i.e. tangram puzzle play). Please find the dynamic version of this image in the supplementary materials.**

is a useful tool that enables running multiple video sources synchronously and insert annotations by setting the gesture taxonomy for an accurate gesture analysis. Given the complexity of gesture coding, we first attained *interpretive validity* (or interpretive understanding) [16], for the authors' account of the meaning of the categories (i.e. goal-directed actions and hand kinematics) and the related sub-categories through the consensus coding. In case of ambiguity or disagreement between the two coders, all four authors took part in the interpretation of each of the categories. This process of coding was repeated several times to ensure that all categories were saturated, meaning that through continuing analysis we formulated descriptions of each sub-categories. The final categories of the gesture taxonomy were revised and finalized as an outcome of these discussions. In this way, the two coders suggested further adding new sub-categories for the main categories i.e. Collect/bring object and Move object sub-categories were added to goal-directed actions, and palm orientations i.e. Forward, Backward, and Two palms facing each other were added under the hand kinematics category.

**4th iteration - Final working through the data and reliability check:** After finalizing the descriptions for each sub-categories, three additional participants' data were coded independently by the second and third authors to check the inter-rater reliability for the coding. We measured the agreement between the two coders for all of the eight codes listed in the goal-directed actions which were derived from an inductive approach in this study. For inter-rater reliability, we calculated intraclass correlation coefficient (ICC) and reported F tests for each ICC: total number of goal-directed actions coded (ICC = .98,  $F(5,6) = 154$ ,  $p < .05$ ), strategy

shift (ICC = .67,  $F(5,6) = 5.12$ ,  $p < .05$ ), align object (ICC = .92,  $F(5,6) = 25.1$ ,  $p < .05$ ), capture / grab object (ICC = .73,  $F(5,6) = 6.5$ ,  $p < .05$ ), collect / bring object (ICC = .99,  $F(5,6) = 215$ ,  $p < .001$ ), leave object (ICC = .92,  $F(5,6) = 26.3$ ,  $p < .05$ ), locate object (ICC = .85,  $F(5,6) = 12$ ,  $p < .05$ ), move object (ICC = .87,  $F(5,6) = 14.9$ ,  $p < .05$ ), and rotate object (ICC = .99,  $F(5,6) = 317$ ,  $p < .001$ ). The inter-rater reliability results were very high for the subjective ratings for goal-directed actions. On that account, we settled on the observable and analyzable categories and sub-categories identifying our gesture taxonomy which we describe in detail below.

## 4 THE MANIPULATIVE GESTURE TAXONOMY FOR VR HAND TRACKING

The taxonomy was developed based on the analysis of children's gestural input for interacting with objects to solve a spatial puzzle task in VR hand tracking. We identified two main categories 1. Goal-directed Actions, 2. Hand Kinematics. In this section, we report on our qualitative results for describing the categories and sub-categories along with quantitative results for frequency in occurring of these variations of actions and kinematics in children's interaction with the objects during the spatial task (i.e. tangram puzzle play).

### 4.1 Goal-directed Actions

While gestures are hand movements, not all hand movements are gestures. Gestures are distinguished from other hand movements in some characteristics as explained by Solomon et al.: Gestures begin from a position of rest, move away from this position, and

| Goal-Directed Actions  |                 | Condition |        | Total |
|------------------------|-----------------|-----------|--------|-------|
|                        |                 | G         | NG     |       |
| strategy shift         | Count           | 30        | 45     | 75    |
|                        | % within row    | 40%       | 60 %   |       |
|                        | % within column | 5%        | 5.5 %  |       |
| align object           | Count           | 78        | 89     | 167   |
|                        | % within row    | 46.7 %    | 53.3 % |       |
|                        | % within column | 12.9 %    | 10.9 % |       |
| capture / grab object  | Count           | 125       | 170    | 295   |
|                        | % within row    | 42.4 %    | 57.6 % |       |
|                        | % within column | 20.6 %    | 20.9 % |       |
| collect / bring object | Count           | 20        | 11     | 31    |
|                        | % within row    | 64.5 %    | 35.5 % |       |
|                        | % within column | 3.3 %     | 1.4 %  |       |
| leave object           | Count           | 129       | 129    | 258   |
|                        | % within row    | 50%       | 50%    |       |
|                        | % within column | 21.3 %    | 15.8 % |       |
| locate object          | Count           | 68        | 113    | 181   |
|                        | % within row    | 37.6 %    | 62.4 % |       |
|                        | % within column | 11.2 %    | 13.9 % |       |
| move object            | Count           | 113       | 95     | 208   |
|                        | % within row    | 54.3 %    | 45.7 % |       |
|                        | % within column | 18.6 %    | 11.7 % |       |
| rotate object          | Count           | 43        | 162    | 205   |
|                        | % within row    | 21%       | 79 %   |       |
|                        | % within column | 7.1 %     | 19.9 % |       |
| Total                  | Count           | 606       | 814    | 1420  |
|                        | % within row    | 42.7 %    | 57.3 % |       |
|                        | % within column | 100%      | 100%   |       |

**Table 1: The total raw number of gestures for each action-directed gesture across Gravity (G) and Non-Gravity (NG) conditions.**

then return to rest. They have a peak structure, also referred to as the stroke, which is generally recognized as a moment of accented movement to denote the function of meaning of a movement; the stroke phase is preceded by a preparation phase and succeeded by a recovery phase in which the hand and arm move back to their rest position. Thus, gestures have a clear beginning and ending. Deriving from this description, we coded the goal-directed actions based on their beginning and ending moments. The rest positions or preparation positions in between actions were not included in our gesture taxonomy.

In total, we coded 1738 manipulative gestures, of which 302 (%17.4) were gestures due to system breakdown and 16 (%0.9) were other gestures (e.g. off-task hand movements and obscure gestures). For final analyses, we discarded the latter two types of gestures and reported goal-directed actions only (N= 1420). Table 1 presents the raw number and percent of each goal-directed action category across Gravity (G) and Non-Gravity (NG) conditions. As can be seen from Table 1, the two most frequent goal-directed actions are Capture / grab object (%21) and Leave object (%18) across all participants and conditions. The goal-directed actions are described below:

1. *Collect / bring object*: This action happens when the users bring one or more objects closer to themselves. This action also

usually comes before grabbing action to prepare the object available for capturing. However, the end goal is not necessarily always capturing. The purpose of this action may also lead to bringing objects closer to the region they work on building the figure or their region of interest. Since the subsequent action varied, we coded this action as a separate sub-category from Capture / grab.

2. *Capture / grab object*: We coded this action when the users start to bring their hands towards the aimed object until they complete to hold the object. This action usually precedes other actions such as move, locate or rotate. Thus, we only coded the action within the time frame that starts with a clear beginning to the end of a mere capturing target.

3. *Move object*: This action refers to lifting and carrying an object from one place to the other, albeit not necessarily to the region of interest where the user works on the figure. The move action stops when the user leaves the object which is coded as a separate action.

4. *Locate object*: This action is coded when the user placed an object within the working area where s/he works on the figure. It refers to an intentional spatial thinking for placing of an object within an estimated region of interest which involves scaling the objects' location in mind first.

5. *Leave object*: This action basically refers to the moment when the user stops an action with the object and releases the object. It is one of the most recurring action after Capture /grab, because all actions end with leaving the object. Also, during some Rotation actions especially rotating in an object with a wider angle, the users use Leave gesture in the middle of the Rotation action to complete the entire rotation of the object to bring it to the desired orientation. It is coded as a separate action because in VR hand tracking systems leaving an object requires the user to spread the fingers more than usual in the physical world.

6. *Rotate object*: This action is at the core of the mental rotation skill which is the physical action for bringing an object to a required orientation. It involves bringing the edges and corners of an object to the position with respect to the other objects in the entire figure. Interestingly, children employed rotation action much more frequently in Non-Gravity condition.

7. *Align object*: This action takes place when the edge of an object is intended to be aligned with the neighboring object. Thus, this action occurs only when at least one neighboring object is present and the action is conducted accordingly.

8. *Strategy shift*: This action shows an intentional change in strategy of manipulating an object. It occurs in various cases such as when the user shifted the position of hand to hold the object, or when changing the mind in positioning or locating of an object, or basically leaving an object to continue with working with another object instead on that point. We also coded the actions under this sub-category when the users change their hand to be able to rotate an object more comfortably or ergonomically.

## 4.2 Hand Kinematics

Hand kinematics include children's hand and finger configurations while interacting with the spatial objects (i.e. tangram pieces). Here we present our results for hand configuration, active hand use, active finger use, and palm orientation respectively in relation to the goal-directed actions to provide an understanding of children's gesture input as a combination of these two dimensions.

*Hand & Finger Configurations*: We measured the raw number and percentages of hand configurations across each goal-directed action and condition (Gravity vs. Non-Gravity) and here we report the two most frequently used hand configurations for each goal-directed action (see Figure 2). Based on the raw numbers and the percentages of each hand configuration within each goal-directed action category, the two most frequent hand configurations were two-finger pinch with fist and two-finger pinch with open palm, respectively in strategy shift (33, %44.6 and 26, %35.1), align object (86, %51.5 and 65, %38.9), capture/grab object (114, %38.9 and 143, %48.8), locate object (79, %43.6 and 72, %39.8), move object (87, %42 and 61, %29.5), and rotate object (72, %35.1 and 89, %43.4). For the leave object, participants most frequently used open-palm only (142, %55.7) and two-finger open (71, %27.8) hand configurations. Lastly, for collect/bring object, the two most frequently used hand configurations were fingertip + open-palm (22, %71) and open-palm only (5, %16.1).

*Active Hand Use*: Across all goal-directed actions, participants tend to use one-hand configurations: either left-hand only (571, %40.2 of all gestures) or right-hand only (795, %56 of all gestures).

Both-hand use was more frequent in rotate object (N= 17) and strategy shift (N= 11) categories, following leave object and move object (N=7 for each) categories. In all these categories, most of the two-hand gesture use consisted of simultaneous use of both hands.

*Active Finger Use*: We also analyzed the number of fingers used for each hand configuration across all goal-directed action categories. Two-finger use was automatically coded for two-finger pinch with fist, two-finger pinch with open-palm, and two-finger open hand configuration categories. For fingertip-only, two-finger use was the most frequent (%45.6) followed by five-finger use in %26.6. Participants used five fingers most frequently with fingertip + open-palm (%92.4), open-palm only (%85.5), and whole-hand grabbing (%90.3) hand configurations.

*Palm Orientation*: Participants used palm-down gestures the most frequently (%48 of all gestures). For palm-orientation, we analyzed hand movement direction across Gravity (G) and Non-Gravity (NG) conditions (see Table 2 in Supplementary Materials). The results show that participants used comparable number of palm-down and palm-up gestures across G and NG conditions. However, participants used more palm-back, palm-front, palm-left, palm-right, and palms facing each other gestures in NG condition compared to G condition.

## 5 DISCUSSION AND FUTURE DIRECTIONS

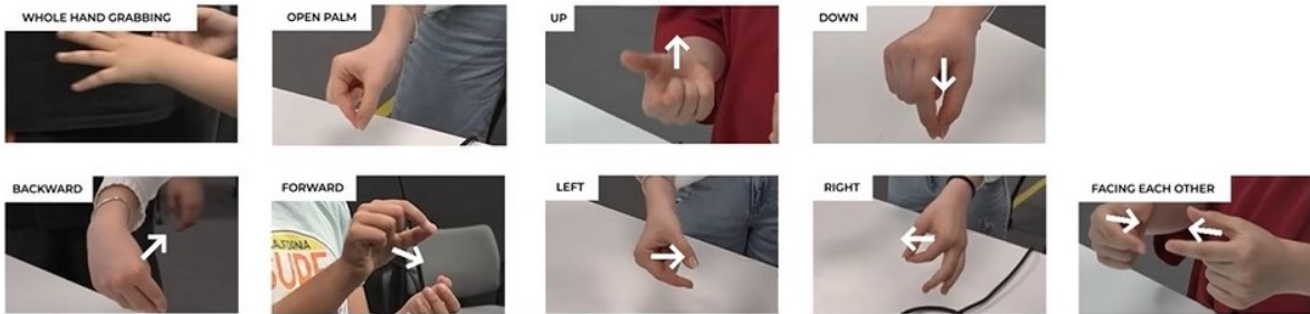
In this paper, we have introduced a manipulative gesture taxonomy for analysing children's object interaction in spatial puzzle play in VR hand tracking system. The taxonomy consists of two broad categories Goal-directed Actions about a mental rotation task and Hand Kinematics as complementary dimensions of gestural input, and we propose that a combined use for analysis would provide a comprehensive approach for further studies in gesture and spatial thought relation as a complex phenomena. The preliminary results of the quantitative analysis also shed light to further directions to research and design of VR hand tracking systems in service of children's object interaction especially for mental rotation skills. Since the focus of this paper was to develop a gesture taxonomy, suggestions for system development remains beyond the scope of this study. However, further studies could use this taxonomy to identify and analyse when and how children intend to use different gestural input to interact with the object with respect to the proposed taxonomy, as well as understanding children's interaction types for improving hand tracking systems for children's hand motion.

One of the most prominent preliminary findings of this study is that children employed rotation action in Non-Gravity condition to a considerably larger extent when compared to Gravity condition. One reason for this is the fact that, non-gravity condition allows more free use of hands, removing the physical contact with the objects. Also, we may discuss hand use in horizontal vs vertical axis. In horizontal axis (palm-up and palm-down), we may not change hand use according to gravity. However, gravity might affect hand configuration on the vertical axis (e.g., palm-left or palm-front). Thus, the position of an object upon Leave object action is not limited with the two axis anymore in the Non-Gravity condition. This allows 360 degrees of freedom for one can manipulate

## FINGER CONFIGURATIONS



## HAND CONFIGURATIONS



**Figure 2: Hand kinematics including hand and finger configurations above and palm orientations below that children used while interacting with the objects during the spatial task (i.e. tangram puzzle play). Please find the dynamic version of this image in the supplementary materials.**

an object, and so for rotating it. Given the direct link with gesture production and spatial thinking for cognitive development, this freedom may create an opportunity for mental rotation skills of children in VR hand tracking. Yet, object interaction in non-gravity may also create its own challenges in Align object action for children. However, we believe Rotation of spatial manipulatives in Non-Gravity context in VR is worth further investigation for its possible benefits to children's spatial thinking practices such as mental rotation, mental visualization, mental orientation, and mental folding. Moreover, future work with a larger sample size will benefit from inferential statistical analysis to observe whether there are any differences across (sub)categories and conditions in statistically significant terms, especially comparing physical and virtual environments for object interaction.

One limitation of this study is that our investigation was limited with tangram objects which offer a triangular and relatively flat surface when compared to other spatial manipulatives that offer curvilinear forms such as Fröbel Gifts, or 3 dimensional shapes such as wooden blocks or Lego bricks. Still, we expect the goal-directed actions would not differ fundamentally in those objects, but the hand and finger configurations may show different variation or frequency for those objects as well as diverse interaction modalities for design. Thus, further research may examine the form factors by using this gesture taxonomy and examine the different opportunities and challenges offered by different types of spatial manipulatives for children's spatial skills, or object interaction experiences in general to further test and develop the proposed taxonomy.

This taxonomy is an initial step towards capturing the complexity of gesture input in VR hand tracking systems design and research. Our preliminary work provides a characterization of

two dimensions of gestural input pertaining to mental rotation, providing researchers and designers a starting point for deeper understanding of the topic.

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## REFERENCES

- [1] Dimitra Anastasiou, Valérie Maquil, Eric Ras, and Mehmetcan Fal. 2016. Design implications for a user study on a tangible tabletop. In *Proceedings of the The 15th International Conference on Interaction Design and Children*. 499–505.
- [2] Jakki O Bailey and Jeremy N Bailenson. 2017. Immersive virtual reality and the developing child. In *Cognitive development in digital contexts*. Elsevier, 181–200.
- [3] Gökçe Elif Baykal, I Veryeri Alaca, Asim Evren Yantaç, and Tilbe Gökşun. 2018. A review on complementary natures of tangible user interfaces (TUIs) and early spatial learning. *International journal of child-computer interaction* 16 (2018), 104–113.
- [4] Gökçe Elif Baykal, Ali Leylekoğlu, Can Bora Sezer, and Işıl Oygür. 2022. Studying Children's Manipulative Gestures in Spatial Puzzle Play with VR Hand Tracking: Analysis of Goal-directed Actions. In *Proceedings of the 2022 ACM Symposium on Spatial User Interaction*. 1–2.
- [5] Gökçe Elif Baykal, Maarten Van Mechelen, Tilbe Gökşun, and Asim Evren Yantaç. 2019. Embedded figures in stories (EFIS): A method to observe preschoolers' interactions with spatial manipulatives. *International Journal of Child-Computer Interaction* 21 (2019), 121–129.
- [6] Shimmila Bhowmick, Keyur Sorathia, and Pratul Chandra Kalita. 2021. Understanding Gesture Performance for Object Selection in VR: Classification and

- Taxonomy of Gestures in HCI. In *India HCI 2021*. 90–93.
- [7] Eunjung Choi, Heejin Kim, and Min K Chung. 2014. A taxonomy and notation method for three-dimensional hand gestures. *International Journal of Industrial Ergonomics* 44, 1 (2014), 171–188.
  - [8] Andrew Clark, Anban W Pillay, and Deshendran Moodley. 2020. A system for pose analysis and selection in virtual reality environments. In *Conference of the South African Institute of Computer Scientists and Information Technologists 2020*. 210–216.
  - [9] Stacy B Ehrlich, Susan C Levine, and Susan Goldin-Meadow. 2006. The importance of gesture in children's spatial reasoning. *Developmental psychology* 42, 6 (2006), 1259.
  - [10] Susan Goldin-Meadow. 2015. From action to abstraction: Gesture as a mechanism of change. *Developmental review* 38 (2015), 167–184.
  - [11] Kara M Gregory, An Sook Kim, and Alice Whiren. 2003. The effect of verbal scaffolding on the complexity of preschool children's block constructions. (2003).
  - [12] Celeste Groenewald, Craig Anslow, Junayed Islam, Chris Rooney, Peter J Passmore, and BL Wong. 2016. Understanding 3D mid-air hand gestures with interactive surfaces and displays: a systematic literature review. (2016).
  - [13] Kathy Hirsh-Pasek, Jennifer M Zosh, Helen Shwe Hadani, Roberta M Golinkoff, Kevin Clark, Chip Donohue, and Ellen Wartella. 2022. A whole new world: Education meets the metaverse. *Policy* (2022).
  - [14] Maria Karam et al. 2005. A taxonomy of gestures in human computer interactions. (2005).
  - [15] The Language Archive Max Planck Institute for Psycholinguistics. 2022. ELAN (Version 6.4). (2022). <https://archive.mpi.nl/ta/elan>
  - [16] Joseph Maxwell. 1992. Understanding and Validity in Qualitative Research. *Harvard Educational Review* 62 (01 1992), 279–300.
  - [17] Philipp Mayring. 2000. Qualitative Content Analysis. *Forum Qualitative Sozialforschung / Forum: Qualitative Social Research [On-line Journal]*, <http://qualitative-research.net/fqs/fqs-e/2-00inhalt-e.htm> 1 (06 2000).
  - [18] Joan McComas, Jayne Pivik, and Marc Laflamme. 1998. Children's transfer of spatial learning from virtual reality to real environments. *CyberPsychology & Behavior* 1, 2 (1998), 121–128.
  - [19] Jess McIntosh, Charlie McNeill, Mike Fraser, Frederic Kerber, Markus Löchtefeld, and Antonio Krüger. 2016. EMPress: Practical hand gesture classification with wrist-mounted EMG and pressure sensing. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 2332–2342.
  - [20] D McNeill. 1992. *Hand and Mind: What Gestures Reveal About Thought*. Chicago: Univ.
  - [21] Nora S Newcombe and Thomas F Shipley. 2015. Thinking about spatial thinking: New typology, new assessments. In *Studying visual and spatial reasoning for design creativity*. Springer, 179–192.
  - [22] Vladimir I Pavlovic, Rajeev Sharma, and Thomas S. Huang. 1997. Visual interpretation of hand gestures for human-computer interaction: A review. *IEEE Transactions on pattern analysis and machine intelligence* 19, 7 (1997), 677–695.
  - [23] Francis KH Quek. 1994. Toward a vision-based hand gesture interface. In *Virtual reality software and technology*. World Scientific, 17–31.
  - [24] Amber Solomon, Mark Guzdial, Betsy DiSalvo, and Ben Rydal Shapiro. 2018. Applying a gesture taxonomy to introductory computing concepts. In *Proceedings of the 2018 ACM Conference on International Computing Education Research*. 250–257.
  - [25] David H Uttal, David I Miller, and Nora S Newcombe. 2013. Exploring and enhancing spatial thinking: Links to achievement in science, technology, engineering, and mathematics? *Current Directions in Psychological Science* 22, 5 (2013), 367–373.
  - [26] Dylan Yamada-Rice, Faisal Mushtaq, Adam Woodgate, D Bosmans, A Douthwaite, I Douthwaite, W Harris, R Holt, D Kleeman, J Marsh, et al. 2017. Children and virtual reality: Emerging possibilities and challenges. (2017).
  - [27] Ying Yin. 2012. A hierarchical approach to continuous gesture analysis for natural multi-modal interaction. In *Proceedings of the 14th ACM international conference on Multimodal interaction*. 357–360.